

Supported OS

- Windows 10 Pro/Enterprise 64-bit (recommended)
- Windows 11 Pro/Enterprise 64-bit
- Windows Server 2019/2022 (optional)

Minimum Hardware

- Dual-core 2.0 GHz CPU
- 4 GB RAM (8 GB recommended)
- 20 GB free disk
- 100 Mbps LAN



Balaji FeeHub

Fee Management System

Client PCs

- Windows 10/11 64-bit
- Chrome OR Microsoft Edge
- LAN connection to Main Server
- NO Python, Node.js, or MongoDB required

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Installation Manual

Version 1.0

Offline · LAN-based · Auditable · Precise

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2. Static LAN IP

Assign a static IPv4 to the Main Server

- Open Control Panel → Network → Change Adapter Settings
- Right-click Ethernet → Properties → IPv4 → Properties
- IP: 192.168.1.10 · Subnet: 255.255.255.0 · Gateway: 192.168.1.1
- The installer detects this IP and writes it into frontend/.env — no manual edit needed

3. Firewall

Ports opened by install-main-server.bat

- TCP 3000 — Application (opened)
- TCP 8001 — Backend API (opened)
- TCP 27017 — MongoDB (kept private, bound to 127.0.0.1)

Windows Defender

- The installer runs `netsh advfirewall firewall add rule ...` automatically
- You do NOT need to click through any Firewall dialog on the Main Server

4. MongoDB

Bundled MSI

- The FINAL ZIP ships with the official MongoDB Community MSI under 05-services/
- The installer runs it in silent mode: ``msiexec /i mongodb-*.msi ADDLOCAL=ServerNoService /qn /norestart``

Configuration

- MongoDB binds to 127.0.0.1 only (bindIp: 127.0.0.1 in mongod.cfg)
- Clients never talk to MongoDB directly — every request goes through the backend on port 8001

Service

- Registered as ``BalajiFeeHub-Mongo``, auto-start
- Data dir: `C:\balaji-fee\mongodb\data`
- Log dir: `C:\balaji-fee\mongodb\logs`

5. Backend & Frontend Services

Backend

- `BalajiFeeHub-Backend` — unicorn on port 8001
- Depends on Mongo service, auto-restart via NSSM
- Log rotation at 20 MB, 5 files

Frontend

- `BalajiFeeHub-Frontend` — Python built-in `http.server` on port 3000
- Serves the prebuilt React static site from frontend/build/
- No Node/npm required on the Main Server

Auto-start

- All three services set to Auto — Windows reboots do NOT require manual intervention

6. First Login

Default admin

- admin@balajiconvent.in / ChangeMeOnFirstLogin@2026
- Change password IMMEDIATELY from Menu → Users
- Factory Reset PIN default: 2580 — change from Administration → Factory Reset → Change PIN

7. Client PC Installation

Zero-edit installer

- Copy 02-install-client-pc/ to the client PC (from Main Server share or USB)
- Right-click install-client-pc.bat → Run as administrator
- The script auto-discovers the Main Server on the LAN (10-second parallel scan)
- If discovery fails, admin enters the IP manually
- Desktop shortcut + Start-Menu shortcut are created
- Browser opens automatically

8. Multiple Cashier PCs

Concurrent access

- The backend supports 100+ concurrent requests (load-tested to 120 req/s)
- Each cashier logs in with their own credentials on their own PC
- Every write is recorded in the audit log with cashier name + timestamp + IP

9. Printer Setup

Any Windows-supported printer

- Add via Control Panel → Printers → set as default
- Print a Windows test page first
- In the receipt view, click Print — the browser dialog opens with the default printer preselected

Paper sizes

- A5 (default), A4, A4-Landscape, Legal, Letter, Thermal 80 mm — configurable per receipt type

10. Backup & Restore

Automatic

- Daily backup runs at 02:00 · last 30 backups retained · integrity SHA-256

Manual

- Administration → Backups → Create Backup Now (Admin PIN)

Restore

- Administration → Backups → Restore (requires Admin PIN + password re-verify)

11. Uninstall & Repair

Uninstall

- 01-install-main-server/uninstall.bat (services removed; data + backups kept)

Repair

- 01-install-main-server/repair-installation.bat (re-copies source, re-runs pip install)

Client Uninstall

- 02-install-client-pc/uninstall-client-pc.bat

12. Troubleshooting

Client cannot connect

- ping 192.168.1.10 from cmd; confirm firewall rules; restart services in services.msc

Login fails

- Reset password from Menu → Users (as another admin) or from install-main-server .env

Update failed

- System auto-rolled back; check Software Updates → History for the failed step

Printer not working

- Set as default in Windows; test with a Windows test page

13. Not Yet Tested on Clean Windows PC

Developer transparency

- This ZIP was assembled in a Linux container.
- Every .bat script was hand-written to Windows conventions and reviewed against the 20-point checklist.
- Clean-machine acceptance tests (Windows 10 + Windows 11) MUST still be executed by the school before daily use.
- If preflight.bat reports any BLOCK line on your Main Server, share it and we will patch immediately.